

Flexible work practices and the firm's need for external orientation

An empirical study of SMEs

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Abstract

Purpose – Literature on small and medium enterprises (SMEs) has so far produced limited evidence on how these firms pursue their organizational flexibility with information and communication technology (ICT) and ad hoc work practices. The purpose of this paper is to contribute to the extant literature by focusing on how SMEs use flexible work practices that provide latitude with respect to when employees work, where they work and via which communication medium. Specifically, the authors analyze how such practices are related to the conditions that SMEs face in reference to their competitive environment and their patterns of ICT usage.

Design/methodology/approach – A survey was conducted on 304 Italian SMEs, with the aim of identifying the contextual dimensions where flexible work is chosen and the different typologies of flexible work implemented by companies.

Findings – Flexible work in SMEs is chosen for different reasons associated to different conditions in the competitive environments and in ICT usage where SMEs operate. In general, SMEs use flexible work when they are more capable of improving their external orientation toward suppliers, customers, and the entrance in new markets with ICT. This duality is more likely in the competitive environments where external orientation and information processing is more needed, namely, environments that are uncertain and complex for product and breadth of the geographical complexity (scope) covered.

Research limitations/implications – In this paper, the authors offer an analysis on the contextual characterizations of flexible work practices. Future studies should disentangle more in depth the ways these characterizations are related to different ICT usages.

Practical implications – In uncertain and complex environments, SMEs should increase their external information processing with ICT and organizational practices that support the latitude of employees involved in boundary spanning with respect to where, when, and how they work.

Originality/value – This paper offers an interpretation of flexible work as an organizational mechanism used to cope with uncertain and complex environments where more external orientation is needed. This paper also shows that there are four different typologies of flexible work implemented by companies, namely, flexible work for cheaper input costs, flexible work for operational drivers, flexible work for strategic drivers, and flexible work for individual motivations, and that in some cases the conditions under which they are chosen are different.

Keywords Environmental uncertainty, Small and medium enterprises, Flexible work practices, ICT-based external orientation

Paper type Research paper

Introduction

Work is becoming more flexible as developments in technologies allow employees to work “any time and any place,” thereby enabling new forms of temporal and spatial flexibility. Although the diffusion of new technologies, like tablets and smartphones, and the rise of the cloud computing have revamped the interest in the usage patterns of flexible work forms, there is still a limited understanding on the organizational context that lead firms to provide latitude with respect to when employees work (temporal flexibility), where they work



(spatial flexibility) and via which communication medium (smart phone, e-mail, and videoconference). The lack of a unified and comprehensive framework over the reasons why firms use flexible work is related to the fact that flexible work encompasses a large variety of work practices, such as home-based work (Wheatley, 2012), mobile work (Raguseo *et al.*, 2015), flexible work (Hoeven and Zoonen, 2015), smart work (Raguseo *et al.*, 2016; Kim and Oh, 2015), and dispersed work (Corso *et al.*, 2006). Such concepts depict the evolution in the use of flexible work arrangements that has been taking place over the last 40 years, they have in common a temporal or a spatial flexibility of work, but they also reflect a heterogeneous set of reasons at the origin of flexible work arrangements. Such reasons have been commonly identified in goals such as reducing real estate costs (Mello, 2007), combining proximity to clients with an efficient use of employees' working capacity (Rapp *et al.*, 2012), favoring workers' life-work balance (Pulido and López, 2005), and facilitating communication flows among employees (Bélanger and Allport, 2008). However, such reasons do not shed light over the link between firm's competitive strategies and the way flexible forms of work are used.

Flexible work arrangements can be one of the practices that firms use to support their flexibility, namely, the capacity to adapt at a limited cost their competitive actions (e.g. product mix, new products or new product features, and product volumes) to what customers need and to new market opportunities (Martinez-Sanchez *et al.*, 2008). In this paper, we try to bridge the gap on the link between firm's strategy and flexible forms of work by analyzing under which conditions of their competitive environment and their strategy small and medium enterprises (SMEs) recur to flexible forms of work, and by identifying the different typologies of flexible work chosen by SMEs.

Our interest on SMEs is motivated by the fact that SMEs may develop models of flexible work that differ significantly from those followed by larger organizations, which have been object of the majority of studies on flexible work so far. Compared to larger enterprises SMEs face different conditions in relation to their labor contractual framework, formalization of human resource management practices, degree of unionization, and geographical scope of their operating units. Furthermore, flexible structures are a source of SMEs' strengths over large firms (Terziovski, 2010).

To analyze the linkage between SMEs' strategic positioning and the use of flexible forms of work we explore how SMEs use their information and communication technologies (ICT) in support of their competitive strategies. More specifically, we assess whether the motivations and the mechanisms at the origin of flexible work can lie in firms' need to improve their external orientation (Chung *et al.*, 2014). An externally oriented firm prioritizes the coordination with partners (e.g. distributors), customers and suppliers, the reconfiguration of their network of partners and their product development capabilities over the objectives of internal efficiency. Such strategic posture can be typical of competitive environments characterized by a high level of uncertainty and complexity in technological and market conditions. Contingency theory in organizational studies illustrates that complexity and uncertainty require firms to increase their information processing capacity and the use of boundary-spanning roles (Leifer and Delbecq, 1978). Flexible forms of work pursued through the spatial mobility of workers can be a mechanism that SMEs use to cope with their competitive environment and with the need to have a high degree of coordination with customers and suppliers. In other words, flexible forms of work can allow SMEs to cope with the need of external orientation without reliance on hierarchical and bureaucratic organizational forms.

Based on these considerations, this paper contributes to the existing literature in two ways. First, in an innovation diffusion perspective, the pattern of the diffusion of flexible forms work in SMEs are figured out. In this regard, our results show that the conditions that lead SMEs to opt for flexible forms of work are the environmental uncertainty, and the use

of ICT to support firm's external orientation and to manage a broad geographical scope. Second, with a system use perspective, we shed light on how the above mentioned conditions of SMEs' strategic posture and competitive environment are related to the patterns of use of flexible work arrangements. We do this by focusing on how often firms use flexible work arrangements, the frequency of their use, and the percentage of workers that are involved in flexible work. We thus describe some dimensions of flexible work as an organizational practice (Pentland and Feldman, 2005), like its recurrence in the organizational procedures, the frequency of repetition, and its nature as collective phenomena (Beckman *et al.*, 2004). For achieving these objectives, four hypotheses were tested on a sample of 304 Italian SMEs.

Theoretical background

Referring to organizational studies on flexible organizations (e.g. Volberda, 1998), this section illustrates the concept of flexible work in relation to its role in supporting a firm's external orientation as a way to manage flexibility, as the capability to cope with uncertainty and rapid environmental changes. The general picture emerging from this analysis is that flexible work and ICT, supporting communication and coordination across the firm boundaries, can be levers that SMEs use to increase their external orientation in environments with greater uncertainty and complexity.

Flexible work: definition, cases, and drivers

We define flexible forms of work as the usage of flexible technologies to accomplish tasks across locational, temporal, and contextual boundaries (Martins *et al.*, 2004). Flexible work may be conducted for a variety of reasons and can be described through a spatial, a temporal, and a contextual dimension. The spatial dimension characterizes where workers perform their job when they are "on the move." The contextual dimension characterizes why workers perform their jobs outside the firm's premise and on the move. This can be related to a functional need of the enterprise (e.g. an area manager in the sales function or a technician for a firm in the industrial machinery industry) or it may be related to employees' individual requests. The temporal dimension characterizes the time frame when employees work in mobility (flexible work may occur with a given periodicity or occasionally).

The literature highlights that most forms of flexible work that occur outside the firm may find their reason in the firm's need for flexibility. Flexible work may address a firm's need to cover a broad geographical scope with limited investments in fixed assets, such as in the case of sales representatives who are required to develop business opportunities abroad (Rapp *et al.*, 2012). In a similar way, for some jobs, mobility may address the firm's need to provide technical assistance to sales or after-sales processes (Corso *et al.*, 2006), such as in the case of installers and maintenance specialists in sectors like industrial and office machineries or electricity distribution. In some firms, flexible work may follow the need to dedicate some human resources in their product engineering area to coordination with partners in product development activities, or marketing specialists in their coordination with distributors to adjust the product mix to changes in the demand.

The forms of flexible work illustrated above may represent a mean through which SMEs deploy some key human resources in the exploration of new markets and technological opportunities, such as entry into a foreign market and the absorption of external market or technical knowledge from market or technology partners (Foss *et al.*, 2013). From a strategic perspective, flexible work based on more direct relationships between the firm and its partners may allow SMEs a disintermediation from third parties such as dealers (e.g. export intermediaries) in their access to market knowledge and to the caption of weak signals of change in market and technological conditions.

Another driver of flexible work can be traced back to a firm's need to recombine its resources rapidly in response to changes in the external environment. From this point of view, flexible work can reflect an attempt to gain agility with contingent employees working remotely (Sahaym *et al.*, 2007). In this case, flexible work can be a way to bring new knowledge to a firm without incurring some of the costs associated with direct employment. Firms can implement major changes rapidly because they can hire in new skills quickly and they can shed activities with fewer encumbrances (Kakihara and Sørensen, 2004). Based on this perspective, the increased level of spatial mobility enabled by ICT can lead to transformations in industry and value chain structures and to the emergence of new business frameworks (Lyytinen and Yoo, 2002). For SMEs, this implies that the control enabled by ICT in flexible work might allow new ways to govern business relationships in their value chains and in gaining the organizational agility to pursue business opportunities. Concretely, this may take the form of few flexible workers that develop business opportunities abroad in lieu of export intermediaries, or the firm's capability to take part in product development projects with external partners due to employees that, with a certain periodicity, work remotely in the partners' premises through ICT that support the coordination among the firm and its partners.

The role of ICT in supporting flexible work

ICT may support flexible work in three ways. The first mechanism relates to the reduction of communication costs and to its consequent effect of coordination and collaboration across and within firms. For example, electronic knowledge repositories support knowledge storage and sharing and may thus facilitate coordination within a team of dispersed workers. Enterprise 2.0 technologies (as instant messaging tools, social networks, and online forums and wikis) offer free and easy platforms for communication and interaction. Their use in organizational routines is facilitated by the fact that these technologies are often consolidated in the individual routines of employees' personal lives thanks to the diffusion of tablets and smartphones (McAfee, 2006).

The second mechanism reflects the role that enterprise information systems such as enterprise resource planning (ERP) systems, customer relationship management (CRM), and supply chain management (SCM) tools have in reducing information processing acquisition costs within the enterprise (Yuan *et al.*, 2010) and in imposing standardized and codified work structures on business processes (Neirotti *et al.*, 2011). These technologies allow firms to codify and standardize business processes that cut across organizational groups and operational units (Davenport, 1998). The diffusion of cloud and web-based architectures in ERP and CRM allow firms to give remote access to business processes also through devices such as smartphones and tablets. In this regard, the deployment of flexible work practices exhibits the same technological preconditions of job disaggregation and offshoring trends (Apte and Mason, 1995).

The third mechanism regards the way ICT supports decision making in organization and it can be manifested in different ways. First, ICT can support the introduction of a greater formalization in decision making through the introduction of structured workflow logics. In this regard, web-enabled group decision support systems combine communication, computing, and decision support technologies to facilitate unstructured problem solving and may thus support coordination and decision-making processes across virtual teams of geographically dispersed workers (Majchrzak and Malhotra, 2003). Enterprise systems facilitate flexible working by making decentralization of decision making more efficient. For example, CRM systems can help firms being rapid and effective in responding to some customers' highly specific needs. Whereas enterprise systems may lead to greater delegation of control and decision-making power to flexible workers, some Enterprise 2.0 technologies – by supporting internal networking and reducing communication

costs – may instead favor a trend of centralization of decision-making power. This trend may support agility in the same way of what occurs for the delegation of decision-making power in the day-by-day of flexible workers’ tasks, as centralization of decision-making power may allow firms for a more rapid “management by exception.” Specifically, using Enterprise 2.0 technologies to push up information in the hierarchy can allow flexible workers to rely more on managers for decision making in front of some unforeseen cases. This is especially important for SMEs, given their attitude to centralize many decisions in the hands of the owner but can be important for those type of expertise centralized in particular areas of the organization and cannot be distributed efficiently. For example, in a firm that manufactures molding systems, a sales representative may more easily push up to the engineering department problems that arise in the installation phase of customized machinery. The engineering department can rapidly shoot the troubles and solve the problem. In a similar way, by better interacting with the engineering department, sales representative may more rapidly assess the feasibility and quote an appropriate price for a change request that a customer requires during an advanced phase of the design process of new mold or new machinery.

In sum, ICT and flexible work practices can be related mechanisms through which SMEs can pursue a greater external orientation without the need to rely on bureaucratic and hierarchical configurations.

Hypothesis development

The linkages investigated in this study are represented in Figure 1. The strategic posture in ICT usage can be reflected in two distinct constructs. ICT-based external orientation reflects the way that ICT – by supporting transaction and information processing – allows a firm to manage their boundary-spanning processes, such as product development, logistics, and CRM. The second construct refers to the extent of Enterprise 2.0 technologies usage that firms may use to communicate and collaborate are outside the formalized framework of a given workflow or business process.

The relationship between the strategic posture in ICT usage, environmental conditions, and use of flexible work practices could be intertwined through a bi-directional causality. This is typical of the co-evolutionary phenomena that firms undertake to adapt their organizational forms to changes in the environment (Lewin and Volberda, 1999). This means, for example, that a greater external orientation supported by ICT can lead firms to use flexible workers as a better means to gather and process external information and that – in the same time – the use of flexible work practices can improve external orientation facilitating

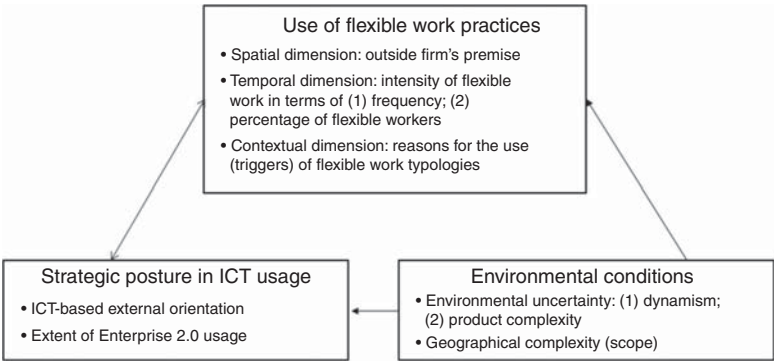


Figure 1.
Conceptual framework

coordination and collaboration with partners and customers. Furthermore, a greater use of flexible work practices can eventually allow a firm to deal with a spreading geographical complexity and with an increasing complexity in the external environment.

Strategic posture in ICT usage

ICT-based external orientation. The concept of ICT-based external orientation refers to a firm's use of information systems to facilitate and enhance its coordination with customers and suppliers and to exploit new strategic opportunities that can be available in the external environment (Sambamurthy *et al.*, 2003). Therefore, it refers to the use of ICT in those departments that most closely interact with the external environment (i.e. sales, marketing, logistics, and product development). Firms that are externally oriented in their ICT use outcomes might be able to respond timely to changes in markets and in customers' needs and to enter into new market segments to exploit new business opportunities in rapid times at a limited cost.

We expect that firms that have developed external orientation through ICT may be more likely to adopt flexible work practices because they are more active in collaboration with supply chain partners, and the use of flexible workers can facilitate collaboration and inter-organizational learning. Inter-firm collaboration requires joint decision making and richer communication based on social interaction between parties. The spatial mobility of employees may provide an avenue for the sharing of richer information through socialization and articulation (Nonaka and Takeuchi, 1995). In this way, in inter-firm relationships, flexible workers may have a role complementary to ICT. Specifically, the role of ICT in inter-organizational relationships is to enhance information exchange by creating process modularity and standard electronic interfaces in supply chain architectures. The supply chain architectures enabled by ICT allow the exchange of information that is very generic and tied to operational processes, but it provides little support for the transfer of idiosyncratic and privileged information that is needed for strategic direction and innovation (Uzzi and Lancaster, 2003). Knowledge creation and innovation through inter-organizational relationships occur when the transactions are embedded in social attachment between parties and in strong inter-firm ties that can be effective for transfer of complex knowledge. Coherent with this view, more innovative firms usually opt for organizational structures with decentralized decision making to better access and process external knowledge during the process of exploiting new strategic opportunities and to respond to external signals (Foss *et al.*, 2013). In SMEs engaged in product innovation with their partners, decentralization can take the form of the spatial mobility of some employees that work within their customers' operational units for some period of time and are actively engaged in the decision-making processes during the product development cycle. These forms of mobility may be driven by the fact that physical distance hinders coordination and knowledge transfer in product design and engineering activities (Allen, 1977). In other cases, decentralization may take the form of a greater empowerment for dispersed sales employees in deciding product configurations and the related pricing or for technical workers engaged in the after-sales services for products with a high degree of complexity, such as industrial machineries. In these cases, the codification of business processes through ERP or CRM systems allows dispersed workers to access the technical and market knowledge that is required to accomplish their tasks with autonomy.

Overall, when SMEs develop a high external orientation through ICT, the need for the mobility of some key roles in the organization can arise in more transactional processes for those employees involved in operational activities (e.g. in sales or customer services for managing customer relationships more promptly) or in innovation endeavors for technical specialists (e.g. an engineer involved in joint product development with supply

chain partners). For externally oriented SMEs, enterprise systems such as ERP and CRM and the tools supporting knowledge management and collaboration in product development processes can provide these employees with the information infrastructure and the workflow structure required to accomplish their tasks, aiding in the coordination and management of dependencies across processes and spatially dispersed workers. For flexible workers involved in more transactional processes, enterprise systems can also produce a decentralization of decision-making rights while simultaneously limiting their individual discretion through *ex ante* compliance with the business rules defining their tasks (McAfee, 2006):

H1. The higher the ICT-based external orientation of SMEs, the greater is their use of flexible work practices.

Extent of Enterprise 2.0 usage. The term Enterprise 2.0 refers to those technologies, such as wikis, social media, and instant messaging tools, which can support lateral communication and collaboration (McAfee, 2006). Most of these technologies are free and easy-to-use instruments that have a broad diffusion and can thus offer SMEs a viable avenue (in terms of usability and costs) to organize and manage knowledge and to support inter-firm collaborations or intra firm-communication. For example, a dispersed team of salespeople can organize a group on a social network such as LinkedIn or Facebook to share experience and knowledge specific to their task environment or to interact with customers to assess their unknown needs.

Compared to CRM and SCM systems, Enterprise 2.0 technologies offer a more effective way of “hardwiring” weak ties in lateral and inter-organizational communication (Schneckenberg, 2009). These mechanisms can allow firms to support the whole knowledge management cycle, from knowledge creation to knowledge storage and sharing.

Therefore, the broader the use of Enterprise 2.0 technologies in a firm’s functional domains (e.g. in communication with suppliers and customers or between employees in the operational lines and specialists in technical staffs), the greater could be the support that the firm can provide to its employees involved in flexible work practices:

H2. The broader the extent of the usage of Enterprise 2.0 tools in SMEs, the greater is their use of flexible work practices.

Environmental conditions

The environmental factors driving flexible work diffusion can be essentially identified in the vast literature on the two main factors that demand firms’ agility and flexibility (Volberda, 1998): the geographical complexity (scope covered by firms’ activities); environment uncertainty (dynamism and product complexity).

Firms with a broader geographical scope are more externally oriented since they must govern a greater variety of products and processes. They thus face a high complexity in the distribution of their products/services and in the acquisition of relevant market information.

Environmental uncertainty can be conceptualized as the interplay between product complexity (i.e. the number of product inputs) and the level of instability in market and input conditions (dynamism). Firms facing a more uncertain environment need to devote more resources to coordination with external partners and to environmental scanning, given the greater complexity of their products and the greater instability in market and technological conditions.

Geographical complexity (scope). The geographical scope refers to the geographical extent of a firm’s operations. Firms facing a broader geographical scope are abler to acquire resources and sell their products and services in international markets, implying that these

firms must access a wide range of information and acquire and absorb context-dependent and idiosyncratic knowledge. Social interaction with foreign partners and customers is thus needed and may occur through dispersed employees who hold boundary-spanning roles. Concretely, in SMEs with a greater geographical scope, boundary-spanning roles search for and assess new business opportunities that can arise in distant markets (Galpin and Sims, 1999) and internalize country-specific knowledge about local relevant market regulations, customer needs, technology standards, etc. We thus expect that SMEs with a broad geographical scope of operations use flexible work practices. Their scale makes economically unsustainable large investments in foreign branch units dedicated to the absorption of external knowledge:

H3. The broader the SMEs' geographical complexity, the greater is the use of flexible work practices.

Environmental uncertainty. Organization theory conceptualizes uncertainty in the firm's environment as the interplay of two dimensions: dynamism and complexity (Galbraith, 1974). Dynamism refers to the degree of instability in an industry in relation to market demand (e.g. changes in customer preference) and technological conditions (e.g. changes in product and process technologies). Product complexity is primarily related to the technical complexity in the number of product components and their interdependencies in the product architecture and thus can be referred to product complexity (Dale Stoel and Muhanna, 2009).

As the amount of uncertainty increases, the need for information processing increases and firms must adopt integrating mechanisms, which increase their information processing capabilities. Uncertain environments require enterprises to deliver offerings at a faster pace and be ready to reinvent their roles in value chain networks, requiring firms to exchange and assimilate a broad range of information from external sources (Galbraith, 1974). Most information exchanges go beyond day-to-day operations and regard long-term dynamics such as market and technology trends, changes in customer preferences, and future product plans. To ensure the quality of information exchanges, SMEs may use flexible work for environmental scanning in the technology portfolios of partners and competitors and for coordination with partners when a revision of product design is needed because of some discontinuities in market demand and technological conditions. Therefore, dynamism and product complexity might influence SMEs' needs for flexible work practices as a decentralization mechanism to improve their external information processing and their coordination with partners for environmental scanning and product innovation activities. Furthermore, past studies have shown that when products become more complex, knowledge about interdependencies becomes more tacit (Cusumano and Selby, 1998) and firms experience a decline in their ability to manage the dependencies among the various product components. An increase in tacit knowledge thus makes it more difficult to effect change when change is needed. This may imply that flexible workers can be needed for integration with external partners as a better way to process and transfer knowledge that is highly tacit and complex:

H4. Environmental uncertainty, seen as the interplay of environmental dynamism and product complexity, is positively associated with the use of flexible work practices in SMEs.

Research methodology

Sample and data collection

The data were collected through a survey sent to SMEs located in the Piedmont region of Italy between February 2010 and June 2010. The firms surveyed had between 10 and 500 employees,

which is a range commonly used to define SMEs (this choice is motivated by the fact that firms with a workforce ranging from 100 to 500 employees have similarities with regard to their hierarchical structure, management style and the types of information systems used, Gray and Mabey, 2005).

Approximately 2,000 companies were selected randomly and contacted by phone to identify the key respondents, who were managers holding the accountabilities for managing information systems investments within their companies since this is the position more knowledgeable in the company for providing information about the possibilities offered by ICT for accessing and coordinating flexible work in SMEs. For companies that did not appoint a specialized ICT management role, we asked for the owner or the CEO.

A total of 304 usable questionnaires were returned, for a redemption rate of 15.20 percent. Testing for non-response bias did not reveal any particular bias related to size, industry, profitability, and key financial and economic indexes.

Measures

Dependent variables. Use of flexible work practices. We used a dichotomic variable to distinguish firms that use a certain type of flexible work practice. We classified a firm as one that had adopted flexible work practices if the following conditions were satisfied:

- (1) The firm has at least one employee who works outside the main premises of the firm at least once a week or when the need occurs. For this criterion, we followed the definition used by (Corso *et al.*, 2006) for geographically dispersed workers, and we included employees who are regularly located outside the firm (e.g. sales agents) and flexible workers who primarily work outside the firm but have their own desk at the firm (e.g. consultants and maintenance technicians).
- (2) During their work outside the firm's premises, flexible workers have access to the firm's enterprise systems through the internet. Checking corporate e-mail remotely was not considered a characterizing attribute. In this way, we were able to capture those cases in which employees can be engaged in business processes remotely, and we could restrict our focus to mobility that involves knowledge workers.

We excluded those situations in which employees work only from their home. In accordance with recent studies on telework (Neirotti *et al.*, 2011), we did not apply any minimum limit to the normal amount of time spent in flexible work, and we thus also included in the generic category of flexible work any employee who works remotely "on-demand" when a particular necessity relating to the firm's business or the worker's private affairs occurs (Kim and Gong, 2016; Jaakson and Kallaste, 2010). We thus focused on those forms of flexible work that occur on a regular basis during the week, also including those forms instrumental to the activities of the firm, even if they do not occur within a predetermined time frame every week.

Contextual dimension of flexible work practices

Flexible work typology. To delineate the typologies of flexible work, we investigated the reasons for the usage of such practices. We identified four types of possible triggers that can motivate the activation of flexible work practices (Table I): an employee's individual reasons; a reduction in the costs of some inputs; operational reasons; and strategic reasons. Those reasons were proposed to firms as being non-mutually exclusive.

Temporal dimension of flexible work

The temporal dimension of flexible work refers to the intensity of usage of this practice. It is represented by the frequency of flexible work and by the percentage of flexible workers.

Flexible work typology (FW)	Flexible work triggers (FWT)	Jobs that can be performed remotely	Main references
FW1: flexible work for individual motivations	FWT 1.1: to meet employees' requests	Any white-collar job in any industry to allow the employee for improving their life-work balance, or to reduce commuting time for going to work	Pulido and López (2005)
FW2: flexible work for cheaper input costs	FWT 2.1: to reduce employees' transfer costs during working time	Technical work in industries such as energy distribution, industrial and office machineries	Neirotti <i>et al.</i> (2011)
	FWT 2.2: to reduce firms' real estate expenses	White-collars who are required to work remotely with a certain periodicity for cost saving purposes such as reduction of office rent, real estate taxes, and property maintenance expenses	Mello (2007)
FW3: flexible work for operational drivers	FWT 3.1: to manage sales, after-sales, and product development activities	Sales and distribution area managers in any industry	Rapp <i>et al.</i> (2012)
	FWT 3.2: for operational continuity reasons related to asset management	Maintenance employees for maintaining firm assets that are spread out over an extended area	Corso <i>et al.</i> (2006)
FW4: flexible work for strategic drivers	FWT 4.1: to access some skills that otherwise the firm could not deploy with standard labor contracts	Professionalized occupations such as designers, software analysts and programmers, marketing analysts, temporary managers, plant managers	Kakihara and Sørensen (2004)
	FWT 4.2: to access skills and convenient wage conditions without any geographical limit	Flexible workers located in satellite centers in rural areas or abroad in developing countries, for taking advantage of the skills of people located everywhere and of cross-countries wages' differences	Bryant (2000)

Table I.
Contextual dimension of flexible work: typologies of flexible work and reasons associated

Frequency of flexible work. We looked at how the frequency of flexible work is institutionalized in firms' standards, rules, and labor contracts. In particular, we distinguished between companies in which some workers are regularly involved in this practice for a predetermined amount of time of at least once a week and other firms in which flexible work mainly occurs occasionally and "on-demand." In this second category, the duration and the frequency of flexible work cannot be formalized a priori based on firm's rules. To exclude cases in which the use of flexible work is more of an exceptional occurrence than a regular practice from this latter category, we set a threshold of at least one day per month for time an employee works in mobility. Specifically, the variable was operationalized with a scale based on the frequency that employees work as flexible workers in a month: occasionally at least one day per month, two or three days a week, or every day.

Percentage of flexible workers. We examined the proportion of the workforce involved in flexible work through the percentage of employees that normally engage in this practice.

Independent variables

Strategic posture in ICT usage. ICT-based external orientation. Our measure of ICT-based external orientation was built on a confirmatory factor analysis (CFA) by modeling a two-second-order construct (Han *et al.*, 2013). The CFA was conducted on scales already tested in previous studies on ICT use in SMEs (Neirotti and Raguseo, 2016), with particular attention to studies on operational agility (Tallon and Pinsonneault, 2011).

The CFA was conducted in two steps. In the first step, the CFA supported the convergent validity of the four factors analyzed: all the estimates for the average variance extracted (AVE) were equal to or higher than 0.50. All scales also exhibited acceptable reliability, achieving Cronbach's α (CA) of at least 0.70 and a composite reliability (CR) more than 0.80. These factors were based on items operationalized by using a five-point Likert scale with responses ranging from "strongly disagree" (−2) to "strongly agree" (+2), in which respondents had to evaluate whether ICT led to a significant impact on a series of items related to the firm's ICT-based external orientation. We asked respondents to base their assessment on the impacts observed over the previous three years. Specifically, the first factor refers to the impact of ICT on new product/service development processes. The second factor reflects how ICT investments support and improve coordination partners in transactional SCM processes and in product design, engineering, and supply activities. The third factor refers to the use of ICT in supporting sales and CRM activities through the enhancement of the knowledge of customers' behavior and service levels in sales and after-sales activities. The fourth factor reflects how firms leverage e-commerce initiatives to increase their revenue volume by entering into new segment markets (in Italy or abroad) or by increasing their penetration of current market segments (Table II).

Indicators	Factor/Item	Mean	SD	Factor loading	CA	CR	AVE
<i>NPD_EO</i>	<i>Improvements in new product development</i>	−0.145	0.783		0.848	0.896	0.683
NPD_EO 1	Growth in the number of new product/services developed	−0.120	1.019	0.764			
NPD_EO 2	Reduction in the failure risks of new products	−0.410	0.875	0.878			
NPD_EO 3	Reduction in time-to-market for new products	−0.260	0.889	0.902			
NPD_EO 4	Improved data management in the product development process	0.150	0.929	0.752			
<i>SCM_EO</i>	<i>Improvements in supply chain management</i>	0.071	0.751		0.907	0.905	0.705
SCM_EO 1	Increased collaboration with suppliers involved in product design and engineering	−0.070	0.868	0.803			
SCM_EO 2	Increased efficiency in the purchasing activities	0.200	0.887	0.832			
SCM_EO 3	Increased quality and/or raw material costs and/or components externally bought	−0.050	0.874	0.872			
SCM_EO 4	Identification of reliable or convenient suppliers	0.170	0.891	0.851			
<i>MKT_EO</i>	<i>Improvements in market knowledge</i>	0.125	0.784		0.852	0.916	0.733
MKT_EO 1	Increased knowledge on customer needs and purchasing habits	0.060	0.937	0.798			
MKT_EO 2	Increased control on sales, included sales agents	0.000	0.950	0.871			
MKT_EO 3	Better support to sales employees	0.100	0.995	0.918			
MKT_EO 4	Improved after-sales services	0.100	0.972	0.834			
<i>ECC_EO</i>	<i>E-commerce competences</i>	−0.307	0.917		0.868	0.912	0.777
ECC_EO 1	Growth in the market share	−0.180	0.998	0.856			
ECC_EO 2	Entry in new market segments	−0.280	1.002	0.943			
ECC_EO 3	Market expansion/entry abroad	−0.450	1.080	0.842			
<i>ICT-based external orientation (EO)</i>		−0.070	0.669		0.848	0.697	0.902
NPD_EO	Improvements in New Product development	−0.145	0.783	0.906			
SCM_EO	Improvements in supply chain management	0.071	0.751	0.853			
MKT_EO	Improvements in market knowledge	0.125	0.784	0.820			
ECC_EO	E-commerce competencies	−0.307	0.917	0.753			

Table II.
Confirmatory
factor analysis

The second step was used to evaluate whether the four factors found in the first step could be grouped into only one factor related to a firm's ICT-based external orientation. In this step, the CFA supported the convergent validity of the four factors analyzed; the second-level factor had an AVE higher than 0.50 and acceptable reliability, as Cronbach's α and the CR were higher than the acceptable thresholds (Table II). For this reason, the second-level factor was named "ICT-based external orientation" and was operationalized by averaging the values of the four previous factors.

Extent of Enterprise 2.0 usage. Enterprise 2.0 usage was operationalized through the use of instant messaging tools (such as chat and systems for video conferences), social networking instruments, and content management systems (such as wikis). This construct was operationalized by counting for each technology its domain of use, distinguishing among three instances in which the tool can support collaboration and communication: with customers, with suppliers, and within the company. In this way, we looked at Enterprise 2.0 usage in relation to its support to a firm's external orientation.

Environmental conditions. Geographical complexity (scope). The geographical complexity of a company was measured by computing the mean of two variables: the extent to which suppliers of product subcomponents are located abroad (geographical breadth of the suppliers), on a four-level scale, and the percentage of revenues made abroad (geographical breadth of the customer base). These two variables were aggregated in a composite variable measuring "geographical scope" after being standardized.

Environmental uncertainty. We kept the two dimensions of uncertainty – dynamism and product complexity – separately, and we tested their individual and joint effects (Dale Stael and Muhanna, 2009).

Environmental dynamism was operationalized as the variability in annual industry sales and measured as the standard error of the regression slope coefficient for annual industry sales divided by the industry mean for the five-year period.

Product complexity was conceptualized as the heterogeneity of inputs. Specifically, for each industry i , the Istat input/output tables were used to calculate the complement to 1 of the concentration of each industry's inputs, measured as $C_i = \sum I_k^2 / (\sum I_k)^2$, where I_k is the volume (in euros) of inputs from industry k .

For both dynamism and product complexity, we ranked the values into two sets (high and low), based on the median value for that characteristic. This choice was mainly motivated by the distribution of the two measures, which resulted non-normal from a Shapiro-Wilks test. Finally, they were standardized for running the regression models.

Control variables. As control variables we used firm size (as the logarithmic form of the number of employees), industry type, and firm revenue's dependence on few customers. SMEs with a low dependence on their top customers could need to adopt flexible work practices largely.

Analyses and results

Use of flexible work practices

Descriptive statistics (Table III) reveal three key findings about the dimensions of flexible work under analysis. First, the diffusion of flexible work practices is still rather limited: only 19.08 percent of the firms' surveyed (58 firms) used flexible work practices. Among these firms, the average proportion of the workforce that participated in this practice amounted to 8.80 percent. The greater part of the firms using flexible work practices (79.31 percent) prevalently adopt on-demand forms of flexible work in which the time spent working outside the firm's premises is not predetermined but occurs with a periodicity of at least one day per month. For 13.79 percent of the firms, the prevalent form of flexible work refers to employees working remotely nearly every day.

Table III.
Descriptive statistics
on flexible work
practices (base: SMEs
that adopt flexible
work practices)

Variable type	No. of SMEs	%	Variable description/FWT	No. of SMEs	%
Flexible work usage			Use of flexible work practices, in any of the possible typologies	58	19.08
Flexible work typology			Reasons explaining flexible work use		
FW1	26	44.88	FWT 1.1	26	44.88
FW2	5	8.62	FWT 2.2	5	8.62
			FWT 2.2	1	1.72
FW3	37	63.79	FWT 3.1	23	39.66
			FWT 3.2	16	27.59
FW4	13	22.41	FWT 4.1	9	15.52
			FWT 4.2	4	6.90
<i>Intensity of use of flexible work</i>					
Frequency and duration of flexible work			Occasionally	46	79.31
			Two or three days a week	4	6.90
			Everyday	8	13.79
Percentage of flexible workers			Mean	na	8.80
			Median	na	5
			Standard deviation	na	7.79
Note: na, not available					

For the majority of firms using flexible work (63.79 percent), flexible work responds to operational drivers, related to sales or after-sales processes (39.66 percent) or to asset management needs (27.59 percent)[1].

For 22.41 percent of the firms using flexible work, flexible work finds motivation in the need to access qualified work and bring new skills into the firm. It is worthwhile noticing that when flexible work originates with the goal of agility in the reconfiguration of competencies, the percentage of flexible workers is lower compared to the other cases of flexible work (equal to 4 percent of the firm's employees).

For 44.88 percent of the firms, motivations for flexible work included individual reasons related to intentions to improve work-life balance. Finally, for 8.62 percent of firms using flexible work, the reason for the use of this practice was due to the intention of reducing input costs.

Determinants of flexible work

Table IV illustrates that SMEs on average show a limited degree of ICT-based external orientation, with a mean for the synthetic index equal to -0.070 on a scale ranging from -2 to $+2$. The low external orientation is mainly attributable to the low impact that ICT has in the product development process (mean of -0.145 for the NPD_EO factor) and in innovations in distribution strategies: for example, few firms have engaged in e-commerce initiatives that allow a penetration of new market segments. In a similar way, a low percentage of firms (36.12 percent) use at least one Enterprise 2.0 tool in their relationships with a customer or a supplier. Therefore, the extent of Enterprise 2.0 tools in supporting firms' external relationships is limited for the majority of SMEs. Firms with an international geographical scope were 9.54 percent of the sample: this percentage refers to companies that have a certain number of suppliers located abroad, and have at least 25 percent of their revenues made from foreign sales.

Overall, these statistics show that the percentage of firms having high levels of external orientation is limited. This appears consistent with the limited diffusion of flexible work practices in the sample.

Table IV.

Independent variables
(descriptive statistics)

Independent variables	Mean	SD	Min.	Max.
<i>Strategic posture in ICT usage</i>				
EO	-0.070	0.669	-2.000	1.920
NPD_EO	-0.145	0.783	-2.000	2.000
SCM_EO	0.071	0.751	-2.000	2.000
MKT_EO	0.125	0.784	-2.000	2.000
ECC_EO	-0.307	0.917	-2.000	2.000
Extent of Enterprise 2.0 usage	0.890	1.279	0.000	8.000
<i>Environmental conditions</i>				
Geographical complexity (scope)	0.000	1.000	-0.860	2.130
Geographical breadth of the suppliers	1.932	0.927	1.000	4.000
Geographical breadth of the customer base	0.127	0.178	0.000	0.500
Dynamism	0.014	0.009	0.003	0.117
Product complexity	0.186	0.081	0.016	0.760
<i>Control variables</i>				
Firm size	3.560	0.970	2.300	6.200
Firm revenue dependence	0.347	0.221	0.150	0.700

Validation of the hypotheses

To explore the drivers of flexible work, we applied regression models on seven types of dependent variables (Table V). Specifically, a Probit regression model (Model 1) was applied to estimate the drivers for the use of flexible work in general. The same model specification was applied to search for the determinants of the four types of motivations explaining the use of flexible work (from Models 2 to 5) listed in Table I.

A second set of models was applied to the subset of the firms in the sample doing flexible work and was aimed at capturing how firms' ICT usage, and their external environment influence the intensity of the use at the firm level of flexible work. This second set of regression analyses included an ordered Probit model for assessing the drivers of flexible work frequency (Model 6) and a Tobit model for the effects explaining the share of the workforce included in this practice (Model 7).

H1 asserted that ICT-based external orientation is associated with a greater firm's use of flexible work. Model 1 showed that firms having a higher ICT-based external orientation show a greater (albeit weakly significant) attitude to use flexible work practices. Specifically, the variable "ICT-based external orientation" has a positive effect on the variable depicting the use of flexible work, which is significant only at the 10 percent level of the *p*-value. However, the level of significance rises to the 5 percent level when we consider the effect of this driver on the use of flexible work for operational reasons (Model 3). Furthermore, Model 7 shows that firms having a higher ICT-based external orientation do not necessarily involve a greater percentage of their workforce in flexible work, but their flexible workers spend a greater percentage of their working time outside the firms' premises (coefficient = 0.685, *p*-value < 5 percent in Model 6). *H1* is thus supported for those forms of flexible work driven by operational reasons.

H2 advanced a positive association between Enterprise 2.0 technologies and flexible work. Model 1 shows that SMEs, with a greater usage of Enterprise 2.0 tools, are more likely to use flexible work practices. The coefficients associated with the variable "Enterprise 2.0 usage" reflect a positive and significant effect in each of the first five models. Furthermore, firms with a greater extent of Enterprise 2.0 usage tend to involve a greater percentage of their workforce in flexible work (Model 7). *H2* is thus supported for every contextual dimension of flexible work.

H3 claimed a positive association between the breadth of SMEs' geographical complexity and their use of flexible work. Model 1 indicates that firms with a broader

Table V.
Regression models on
the drivers of flexible
work practice use

Independent variables	Hypothesis	Dependent variables						
		Use of flexible work practice	For cheaper input costs	Flexible work typology (contextual dimension) For operational drivers	For strategic drivers	For individual motivations	Frequency	Flexible work intensity (temporal dimension) % of flexible workers
Model specification		1	2	3	4	5	6	7
<i>Strategic posture in ICT usage</i>								
ICT-based external orientation	H1	0.357****	0.144	0.541*	0.201	0.102	0.685*	-0.805
Extent of Enterprise 2.0 usage	H2	0.321***	0.650**	0.159*****	0.334***	0.299**	0.142	3.842****
<i>Environmental conditions</i>								
Geographical complexity (scope)	H3	0.245****	0.189	0.020	0.307*	0.522**	-0.849**	-4.220*
Dynamism	H4	0.287*	-0.779**	0.149	0.141	0.203	-0.083	-2.806*****
Product complexity		-0.005	-0.348**	0.097	0.166	-0.125	0.001	-0.690
Dynamism x product complexity		0.358*	-0.352	0.435*	0.226	-0.037	0.267	-4.939*
<i>Control variables</i>								
Firm size		0.343**	0.596*****	0.437***	0.097	0.044	0.166	0.637
Firm revenue dependence		-0.003	-0.004	-0.014*	-0.010	0.009	0.012	-0.126
<i>Fit indexes</i>								
Constant		-2.625***	-5.477**	-2.757***	-2.270**	-2.813***	...	6.343*
Log pseudo likelihood		-101.777	-13.281	-76.178	-32.145	-48.147	-21.076	-124.848
Wald χ^2		40.07***	24.81**	31.31***	35.09***	49.71***	22.61*	3.63**
Pseudo R ²		19.98%	42.63%	19.64%	20.17%	18.80%	18.16%	4.89%
Number of SMEs				304			58	
Type of model				Probit			Ordered Probit	Tobit

Notes: Control variables related to industry type are omitted. *, **, ***, ****, *****p-values < 5, < 1, < 0.1, and < 10 percent levels, respectively

geographical complexity show a higher (albeit weakly significant) likelihood to use some forms of flexible work (Model 1). This positive correlation occurs given employee's individual motivations (Model 5) more than to firms' drivers with an operational focus (Model 3). Furthermore, geographical complexity positively influences the likelihood of flexible work motivated by strategic reasons related to the need to bring new skills into the company (Model 4). Specifically, its coefficient is positive and significant only when flexible work for individual or strategic reasons is considered (Models 4 and 5), whereas it appears non-significant when the flexible work is motivated by reasons related to the firms' operations (Model 3) or to the firms' intent to reduce their input cost (Model 2). The analysis restricted to the firms using flexible work (Models 6 and 7) shows that firms facing a broader geographical complexity tend to involve a lower share of their workforce in these practices (Model 7) and in a more limited amount of their working time (Model 6). It thus seems that for firms facing international markets, flexible work follows needs that occur regularly but with a low periodicity and that are aimed at coordination with foreign suppliers and partners and involve few persons in the organization. *H3* is thus confirmed for what concerns the occurrence of flexible work, but not for the intensity of its use at the firm level.

H4 asserted that in more uncertain environments SMEs are more likely to use flexible work. In this regard model specifications show that firms dealing with a higher degree of dynamism are more likely to adopt flexible work (Model 1) but they involve a lower percentage of their workforce. Product complexity, the other dimension of environmental uncertainty, does not influence the use of flexible work, but it positively moderates the impact of dynamism on the use of flexible work (Model 1). Thus, the environmental conditions where firms are more likely to use flexible work are industries that combine both high dynamism and great product complexity. In these situations, however, firms using flexible work tend to involve a lower extent of their workforce in enterprise mobility. *H4* is thus confirmed for what concerns the occurrence of flexible work, but not for the intensity of its use in firms.

Table VI reports a synthesis of the results in relation to the hypotheses.

Discussion

The analysis of the contextual dimension of flexible work offers a support to the idea that the use of flexible work practices occurs in response to SMEs' need to improve their external orientation and their need to process external information. Specifically, the data show that the use of ICT in support of external orientation toward customers, suppliers, and new markets is often accompanied by the use of flexible work practices motivated by reasons of improving operational agility such as serving customers more promptly, or achieving a better level of integration and collaboration with them in product development (FWT 3.1 and FWT 3.2). Instead, when SMEs cover a broader geographical complexity, their use of flexible work mainly aims to bring new competencies into the organization (FWT 4.1 and FWT 4.2), which is another way through which SMEs may pursue their organizational agility.

Under situations of environmental uncertainty, SMEs tend to use flexible work for operational reasons related to their coordination needs with partners (FWT 3.1 and FWT 3.2), whereas they do not show a higher tendency to use flexible work to reconfigure their competencies (FWT 4.1 and FWT 4.2). The positive correlation between environmental uncertainty and flexible work for operational reasons is coherent with the fact that in contexts of dynamism and product complexity, SMEs need to process a wider range of information (Galbraith, 1974) and to exchange idiosyncratic and tacit knowledge with partners (Malhotra and Majchrzak, 2005). In these settings, flexible work practices can represent the avenue for social attachment and coordination that is required for SMEs to achieve knowledge elaboration and joint decision making.

Essentially, these results highlight that flexible work is associated with two types of organizational flexibility – one at the operational level and a second one at the strategic

Table VI.
Summary of
the results

Independent variables	Dependent variables									
	Flexible work typology (contextual dimension)					Flexible work intensity (temporal dimension)				
	Hypothesis	Use of flexible work	For cheaper input costs	For operational drivers	For strategic drivers	For individual motivations	Frequency	% of flexible workers	Hypothesis supported?	
Model	1	2	3	4	5	6	7			
Strategic posture in ICT usage										
ICT-based external orientation	H1	+	+	+		+				Yes, especially when flexible work follows operational drivers. Furthermore, the higher the ICT-based external orientation, the higher the time spent by flexible workers outside the firm's premise
Extent of Enterprise 2.0 usage	H2	+	+	+	+			+		Yes, a greater usage of Enterprise 2.0 tools supports all the forms of flexible work and it is associated with a greater percentage of employees working remotely
Environmental conditions										
Geographical complexity (scope)	H3	+			+					In part, only when flexible work supports firms' needs of strategic flexibility, and individual motivations
Dynamism × product complexity	H4	+	-					- (*)		In part, product complexity alone does not call for flexible work. Only when product complexity is combined with high dynamism, there is a greater likelihood of flexible work (especially when used for operational reasons). Higher dynamism – combined with higher degree of complexity – leads firms to involve fewer employees in flexible work
Notes: “+” indicates that there is a positive significant effect; “-” indicates that there is a negative significant effect. Blank cells indicate the lack of a significant effect; (*) indicates a weak significant effect (<i>p</i> -value between 5 and 10 percent)										

level – related to the need to bring new competencies in SMEs. Coherently with this idea, in unreported statistics we found that when flexible work originates with the goal of flexibility in the reconfiguration of competencies, the percentage of employees whose work systematically or periodically occurs outside the firm's premises is lower compared to the other flexible work practice explored. This result may be the consequence of the fact that flexible work in these settings involves more professionalized occupations through which SMEs complement their portfolio of competencies. Finally, flexible work can be motivated by a third way to search for organizational flexibility, which is mainly aimed at reducing the cost of inefficiencies in the firm's real estate of their units (FWT 2.2) and in the working time that employees spend traveling for business reasons (FWT 2.1). Not surprisingly, this form has a limited diffusion in SMEs and mainly finds applications in environmental situations of low uncertainty, where in general competition tends to be focused more on costs compared to more dynamic and complex environments.

Firms using flexible work practices show a greater extent of use of Enterprise 2.0 solutions. We may thus conclude that these technologies allow SMEs' technical core, flexible workers, and external partners to be connected and that a broader workforce involvement in flexible work practices is accompanied by a greater number and pervasiveness of these instruments used. This result is coherent with the idea that Enterprise 2.0 technologies offer support for lateral communication and knowledge sharing within and across a firm's organizational boundaries.

Conclusions

Implications for theory

The theoretical insights and the empirical findings discussed in this paper contribute to a more comprehensive view of flexible work that enriches its "classical" vision of as an organizational practice triggered by the need of a dispersed workforce to "stay closer" to customers or to some firm's assets. In this way, we were able to find a link between the use of flexible work practices and SMEs' strategic and the competitive scenario, providing four types of findings, essentially:

- (1) the use of flexible work practices involves a limited percentage of SMEs, and in those organizations that use this practice it involves a limited percentage of employees;
- (2) SMEs use flexible work practices when they are more capable of using ICT and Enterprise 2.0 technologies to improve their external orientation;
- (3) being a means of external orientation, flexible work practices are more frequent in environments where more external orientation is needed, namely, uncertainty industries characterized by more environmental dynamism, product complexity, and broader geographical complexity; and
- (4) the influence of the strategic posture in ICT usage and of environmental conditions is in some cases different according to the typology of flexible work chosen by SMEs: flexible work for cheaper input costs, flexible work for operational drivers, flexible work for strategic drivers, and flexible work for individual motivations.

The rarity of flexible work as an organizational practice in SMEs comes with another rare condition, i.e., the use of ICT and Enterprise 2.0 technologies to enhance their external orientation. Flexible work is thus an organizational practice that is complementary to the usage of ICT in support of market and supply chain relationships.

In highlighting these trends, our study echoes research on the role of ICT in enabling firms' greater reliance on market relationships (Malone, 2004), and is consistent with how scholars viewed ICT in the 1990s: an enabling factor for new forms of industry value chains that represent "an important 'bootstrapping mechanism' of long-predicted shifts from hierarchical to market forms of organizations, at least in professionalized sectors of the

economy” (Pickering and King, 1995, p. 479). In this view, flexible work can be one of the intermediary forms between hierarchical and market-based solutions that can provide an avenue for SMEs’ agility. Flexible workers can be – from an information processing view (Galbraith, 1974) – a bridge between SMEs and their partners and, from a strategic perspective, agile and flexible mechanisms to use the external labor market to deploy specialized competencies when needed.

The view of ICT as a force reshaping work, value chains, and organizations is consistent with our evidence on flexible work and leads the way to conceptualize many types of flexible work practices used in SMEs as one of the elements of new organizational forms based on low bureaucracy and mobility of part of the workforce, aimed at achieving a great organizational flexibility in complex and uncertain environments. One of the elements found in this study that could support this view is the limited diffusion that flexible work practices have among SMEs. Firms using flexible work are likely to operate in dynamic and complex environments where they address a broader geographical complexity and they take advantage of Enterprise 2.0 tools and of the support that ICT can give to their market and supply chain relationships. This support of ICT to external orientation is another rare attribute in SMEs and may thus suggest that these firms prevalently use flexible work practice as one of the micro-foundations of new – and yet scarcely diffused – organizational forms that they develop to address uncertainty and variety in their external environments.

These arguments suggest that by using the traditional theoretical position on flexible work as a workplace innovation that gives employees and firms more flexibility (Chung *et al.*, 2014), we may risk seeing “the trees and not the forest” in the rise of new forms of loosely coupled organizations (Sahaym *et al.*, 2007) or “semi-formal” organizations (Biancani *et al.*, 2014), thereby missing the broader implications that flexible technologies have for the reconfiguration of value chains, firms’ organizational boundaries, and workers’ life in their organizational settings.

Implications for practice

This study shows SMEs’ managers that the technological preconditions needed to deploy flexible work practices are not complex because they consist of a set of ICT tools – like Enterprise 2.0 solutions – that are objects of commoditization and consumerization dynamics. The complexity rather lies in the way firms may design work and relationships with customers and suppliers to allow new forms of organizational agility. This may primarily explain the limited diffusion of flexible work practices in SMEs. Furthermore, SMEs’ managers should be aware of the fact that flexible forms of work can be chosen given different reasons, namely, for incurring on cheaper input costs, for operational drivers, for strategic drivers, or for employees’ motivations. However, they have to take into account that under different conditions each of these flexible work typologies can be chosen. For example, flexible work for strategic drivers is more likely under high geographical complexity, or flexible work for operational drivers is more likely under high levels of dynamism and product complexity.

Under uncertain and complex environments, SMEs should increase their external information processing with ICT and organizational practices that support the latitude of employees involved in boundary spanning with respect to where, when, and how they work.

Future research

Coherent with the idea that flexible work can be part of a more articulated organizational form that SMEs adopt to have a greater external orientation in a more uncertain external environment, future research should assess more in depth the benefits that flexible work can generate, and how firms design information systems, practices, and tasks for dealing with the need of mobility. Furthermore, future research should also focus on investigating flexible work practice in other countries in order to verify the generalizability of the findings.

Note

1. For three of the four typologies of flexible work, firms using them have been identified through a Boolean “OR” operator on the two reasons that define these configurations.

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